

2 **Supporting Information for**

3 **Deconstructing the gender gap in chess ratings**

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8 Supporting text

9 Figs. S1 to S11

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Supporting Information Text

The negative hypergeometric distribution and Knapp's test

Suppose, within a single federation or overall, there are M men, W women, and the K 'th best woman is ranked R overall. In the following discussion, we treat M , W and K as fixed, and R as a random variable.

Under the null hypothesis that the ranking order is random, the probability that R is equal to some value r is given by the probability mass function of the negative hypergeometric distribution (?):

$$\Pr(R = r) = \frac{\binom{r-1}{K-1} \binom{W+M-r}{W-K}}{\binom{W+M}{W}}, \quad [1]$$

provided $K \leq r \leq M + K$. If the observed rank is $R = R^*$, then the probability f of a result equal to or more extreme than that observed is

$$f = \sum_{r=R^*}^{M+K} \Pr(R = r). \quad [2]$$

This is also equivalent to one minus the cumulative distribution function of the negative hypergeometric distribution, evaluated at $r - 1$. Like in the main text, we convert f into a two-sided p -value via the standard transformation

$$p = 2 \min(f, 1 - f). \quad [3]$$

Focusing on the top-ranked woman player, we computed Eq. (3) for each federation as well as for the aggregated global rating list, for all 12 data filters (ranging from including all players to including only active non-junior players rated at least 1600). For example, in the complete rating list including juniors but excluding inactive players, we have $W = 17,286$, $M = 153,454$, and if $K = 1$ then $R^* = 79$ corresponds to Hou Yifan's rank as the top woman player. Evaluating Eq. (3) in this case, $p = 4.8 \times 10^{-4}$. As in our main analyses, the p -values increase with stricter data filters. When excluding junior, inactive, and lower-rated players, we get $p = 0.066$ (when players rated below 1400 are excluded) or $p = 0.080$ (when players rated below 1600 are excluded). All global results can be seen together in Table S1.

Table S1. Results of applying Knapp's rank analysis to global chess rating data. The first three columns are various data filters; K is the rank of the woman player we focus on ($K = 1$ means the top-rated and $K = 10$ the 10th highest-rated woman); the next two columns are the number of women and men in the data after applying the filters; and the last column is the p -value as calculated from Eq. (3). Results for $K = 1$ and $K = 10$ are visually separated with a small vertical gap in the table.

Juniors	Inactives	Rating floor	K	Women	Men	p-value
Yes	No	1000	1	17286	153454	0.0005
Yes	No	1400	1	8293	113076	0.0080
Yes	No	1600	1	5306	86055	0.0188
Yes	Yes	1000	1	36401	307078	0.0017
Yes	Yes	1400	1	20603	233274	0.0097
Yes	Yes	1600	1	13894	181478	0.0192
No	Yes	1000	1	18200	231803	0.0214
No	Yes	1400	1	14638	201651	0.0298
No	Yes	1600	1	11180	164672	0.0388
No	No	1000	1	5826	107497	0.0381
No	No	1400	1	4299	92417	0.0660
No	No	1600	1	3236	73934	0.0804
Yes	No	1000	10	17286	153454	$< 10^{-4}$
Yes	No	1400	10	8293	113076	$< 10^{-4}$
Yes	No	1600	10	5306	86055	$< 10^{-4}$
Yes	Yes	1000	10	36401	307078	$< 10^{-4}$
Yes	Yes	1400	10	20603	233274	$< 10^{-4}$
Yes	Yes	1600	10	13894	181478	$< 10^{-4}$
No	Yes	1000	10	18200	231803	$< 10^{-4}$
No	Yes	1400	10	14638	201651	$< 10^{-4}$
No	Yes	1600	10	11180	164672	$< 10^{-4}$
No	No	1000	10	5826	107497	$< 10^{-4}$
No	No	1400	10	4299	92417	0.0001
No	No	1600	10	3236	73934	0.0002

Still for top-ranked women but disaggregating the data by federations, the results are very similar to what we have obtained for the global top 1 gap in the main text. When retaining all except inactive players, 35 out of the 79 federations with at least 30-30 players of each gender have significantly higher-ranked men, with just one federation (Lithuania) with a significantly

34 higher-ranked top woman than expected by chance. This discrepancy gradually melts away as junior and lower-rated players
 35 are excluded, culminating in no significant results out of 30 federations when players who are inactive, junior, or rated below
 36 1600 are excluded.

Table S2. Results of applying Knapp's rank analysis to chess rating data disaggregated by individual federations. The first four columns are as in Table S1; "Federations" is the number of federations with at least 30-30 players of each gender after applying the data filters; the final two columns are the number of federations out of these in which women, resp. men, are significantly higher-ranked than expected by chance (with a significance threshold of 0.05 and after false discovery rate correction to the p-values of the federations within each row). For women, the three-letter labels of the federations with a significant result are also shown (LTU: Lithuania; SVK: Slovakia).

Juniors	Inactives	Rating floor	K	Federations	Women stronger	Men stronger
Yes	No	1000	1	79	1 (LTU)	35
Yes	No	1400	1	58	1 (LTU)	16
Yes	No	1600	1	45	1 (LTU)	9
Yes	Yes	1000	1	99	1 (LTU)	65
Yes	Yes	1400	1	79	1 (LTU)	38
Yes	Yes	1600	1	65	1 (LTU)	30
No	Yes	1000	1	77	1 (LTU)	39
No	Yes	1400	1	69	1 (LTU)	26
No	Yes	1600	1	62	1 (LTU)	24
No	No	1000	1	46	1 (LTU)	10
No	No	1400	1	40	1 (LTU)	4
No	No	1600	1	30	0	0
Yes	No	1000	10	79	0	54
Yes	No	1400	10	58	0	18
Yes	No	1600	10	45	0	11
Yes	Yes	1000	10	99	0	84
Yes	Yes	1400	10	79	0	56
Yes	Yes	1600	10	65	0	45
No	Yes	1000	10	77	0	57
No	Yes	1400	10	69	0	45
No	Yes	1600	10	62	0	34
No	No	1000	10	46	0	12
No	No	1400	10	40	1 (SVK)	7
No	No	1600	10	30	1 (SVK)	6

37 We have also repeated the same analysis for the 10th highest-rated woman instead of the top-rated one. The results are also
 38 in Tables S1-S2.

39 In summary, Knapp's rank analysis yields results that are consistent with our permutation approach. This gives further
 40 confidence that the results are robust.

41 Global and per-federation permutation results, for all data filters and metrics

42 The best way to view the data is to use the interactive application provided as a supplement. It can be downloaded from
 43 the public GitHub repository <https://github.com/weijima/gender-gap-chess>. To run the application locally, open `/app/app.R` in
 44 RStudio and click on the "Run App" button (for this to work, all other files in the `/app` folder must also be downloaded and
 45 placed in the same subdirectory). This application is also deployed online at <https://dysordys.shinyapps.io/gender-gap-chess/>,
 46 and can simply be used from a browser. It will be maintained and kept there for at least one year after publication.

47 Though not in an interactive form, the global rating data and permutation results are also given here, in Table S3. We do
 48 not do the same for the per-federation analyses, as that would take over 20 pages even in heavily condensed form. Instead,
 49 they are available either from the `/data` folder of the above GitHub repository, or from the interactive application.

50 Supplementary figures

51 Figs. S1-S11 are structured in the same way as Fig. 3 in the main text, but for the other 11 data filters.

Table S3. Results for the global chess rating dataset. The first four columns are the metric and various data filters; “Observed” is the observed value of the metric (men minus women); ‘Permutation’ is the mean plus/minus one standard deviation of the one million permutation samples; and “p-value” is computed to four decimal precision via Eq. (3) (where f is the number of permutation samples that are less than the observed value), with a parenthetical W or M in case the result is significant at the 0.05 level favoring either women (W) or men (M).

Metric	Juniors	Inactives	Rating floor	Observed	Permutation	p-value
Overall mean gap	Yes	No	1000	206.7	-0.0 ± 2.8	$< 10^{-4}$ (M)
Overall mean gap	Yes	No	1400	78.1	0.0 ± 2.9	$< 10^{-4}$ (M)
Overall mean gap	Yes	No	1600	38.9	-0.0 ± 3.0	$< 10^{-4}$ (M)
Overall mean gap	Yes	Yes	1000	169.2	-0.0 ± 1.9	$< 10^{-4}$ (M)
Overall mean gap	Yes	Yes	1400	72.9	-0.0 ± 1.8	$< 10^{-4}$ (M)
Overall mean gap	Yes	Yes	1600	41.1	-0.0 ± 1.8	$< 10^{-4}$ (M)
Overall mean gap	No	Yes	1000	79.2	0.0 ± 2.4	$< 10^{-4}$ (M)
Overall mean gap	No	Yes	1400	45.0	-0.0 ± 2.1	$< 10^{-4}$ (M)
Overall mean gap	No	Yes	1600	29.0	-0.0 ± 2.0	$< 10^{-4}$ (M)
Overall mean gap	No	No	1000	101.0	-0.0 ± 4.3	$< 10^{-4}$ (M)
Overall mean gap	No	No	1400	26.8	0.0 ± 4.0	$< 10^{-4}$ (M)
Overall mean gap	No	No	1600	7.8	0.0 ± 3.8	0.0416 (M)
Overall median gap	Yes	No	1000	280.0	-0.1 ± 4.1	$< 10^{-4}$ (M)
Overall median gap	Yes	No	1400	100.0	-0.1 ± 4.3	$< 10^{-4}$ (M)
Overall median gap	Yes	No	1600	45.5	-0.0 ± 4.2	$< 10^{-4}$ (M)
Overall median gap	Yes	Yes	1000	228.0	0.1 ± 2.9	$< 10^{-4}$ (M)
Overall median gap	Yes	Yes	1400	90.0	-0.0 ± 3.0	$< 10^{-4}$ (M)
Overall median gap	Yes	Yes	1600	42.5	-0.1 ± 2.6	$< 10^{-4}$ (M)
Overall median gap	No	Yes	1000	83.0	-0.1 ± 3.5	$< 10^{-4}$ (M)
Overall median gap	No	Yes	1400	50.0	0.2 ± 3.1	$< 10^{-4}$ (M)
Overall median gap	No	Yes	1600	24.0	0.2 ± 3.0	$< 10^{-4}$ (M)
Overall median gap	No	No	1000	106.0	-0.0 ± 5.7	$< 10^{-4}$ (M)
Overall median gap	No	No	1400	32.0	-0.1 ± 5.7	$< 10^{-4}$ (M)
Overall median gap	No	No	1600	8.0	0.0 ± 5.5	0.1784
Top 10 gap	Yes	No	1000	229.1	88.7 ± 22.3	$< 10^{-4}$ (M)
Top 10 gap	Yes	No	1400	229.1	109.8 ± 23.0	$< 10^{-4}$ (M)
Top 10 gap	Yes	No	1600	229.1	118.7 ± 23.3	$< 10^{-4}$ (M)
Top 10 gap	Yes	Yes	1000	208.7	88.8 ± 22.5	$< 10^{-4}$ (M)
Top 10 gap	Yes	Yes	1400	208.7	102.9 ± 22.7	$< 10^{-4}$ (M)
Top 10 gap	Yes	Yes	1600	208.7	109.8 ± 22.8	$< 10^{-4}$ (M)
Top 10 gap	No	Yes	1000	208.7	110.5 ± 23.1	$< 10^{-4}$ (M)
Top 10 gap	No	Yes	1400	208.7	114.4 ± 23.2	$< 10^{-4}$ (M)
Top 10 gap	No	Yes	1600	208.7	117.9 ± 23.2	$< 10^{-4}$ (M)
Top 10 gap	No	No	1000	229.1	128.0 ± 23.9	$< 10^{-4}$ (M)
Top 10 gap	No	No	1400	229.1	136.4 ± 24.1	0.0000 (M)
Top 10 gap	No	No	1600	229.1	139.7 ± 24.3	0.0002 (M)
Top 1 gap	Yes	No	1000	208.0	87.3 ± 54.3	0.0005 (M)
Top 1 gap	Yes	No	1400	208.0	103.5 ± 53.5	0.0080 (M)
Top 1 gap	Yes	No	1600	208.0	110.3 ± 53.6	0.0187 (M)
Top 1 gap	Yes	Yes	1000	197.0	81.5 ± 53.6	0.0017 (M)
Top 1 gap	Yes	Yes	1400	197.0	92.7 ± 53.3	0.0097 (M)
Top 1 gap	Yes	Yes	1600	197.0	98.4 ± 53.4	0.0193 (M)
Top 1 gap	No	Yes	1000	197.0	97.8 ± 53.9	0.0215 (M)
Top 1 gap	No	Yes	1400	197.0	100.9 ± 54.1	0.0302 (M)
Top 1 gap	No	Yes	1600	197.0	103.6 ± 54.2	0.0386 (M)
Top 1 gap	No	No	1000	208.0	116.4 ± 54.7	0.0383 (M)
Top 1 gap	No	No	1400	208.0	122.9 ± 55.4	0.0657
Top 1 gap	No	No	1600	208.0	125.7 ± 55.7	0.0810
Gap in standard deviation	Yes	No	1000	15.5	0.0 ± 1.6	$< 10^{-4}$ (M)
Gap in standard deviation	Yes	No	1400	11.1	0.0 ± 1.9	$< 10^{-4}$ (M)
Gap in standard deviation	Yes	No	1600	12.4	0.0 ± 2.1	$< 10^{-4}$ (M)
Gap in standard deviation	Yes	Yes	1000	6.8	0.0 ± 1.0	$< 10^{-4}$ (M)
Gap in standard deviation	Yes	Yes	1400	14.3	0.0 ± 1.1	$< 10^{-4}$ (M)
Gap in standard deviation	Yes	Yes	1600	18.6	0.0 ± 1.2	$< 10^{-4}$ (M)
Gap in standard deviation	No	Yes	1000	-5.0	0.0 ± 1.5	0.0006 (W)
Gap in standard deviation	No	Yes	1400	12.5	0.0 ± 1.3	$< 10^{-4}$ (M)
Gap in standard deviation	No	Yes	1600	18.4	-0.0 ± 1.3	$< 10^{-4}$ (M)
Gap in standard deviation	No	No	1000	-32.7	0.0 ± 2.7	$< 10^{-4}$ (W)
Gap in standard deviation	No	No	1400	-0.5	0.0 ± 2.5	0.8339
Gap in standard deviation	No	No	1600	6.0	0.0 ± 2.7	0.0263 (M)

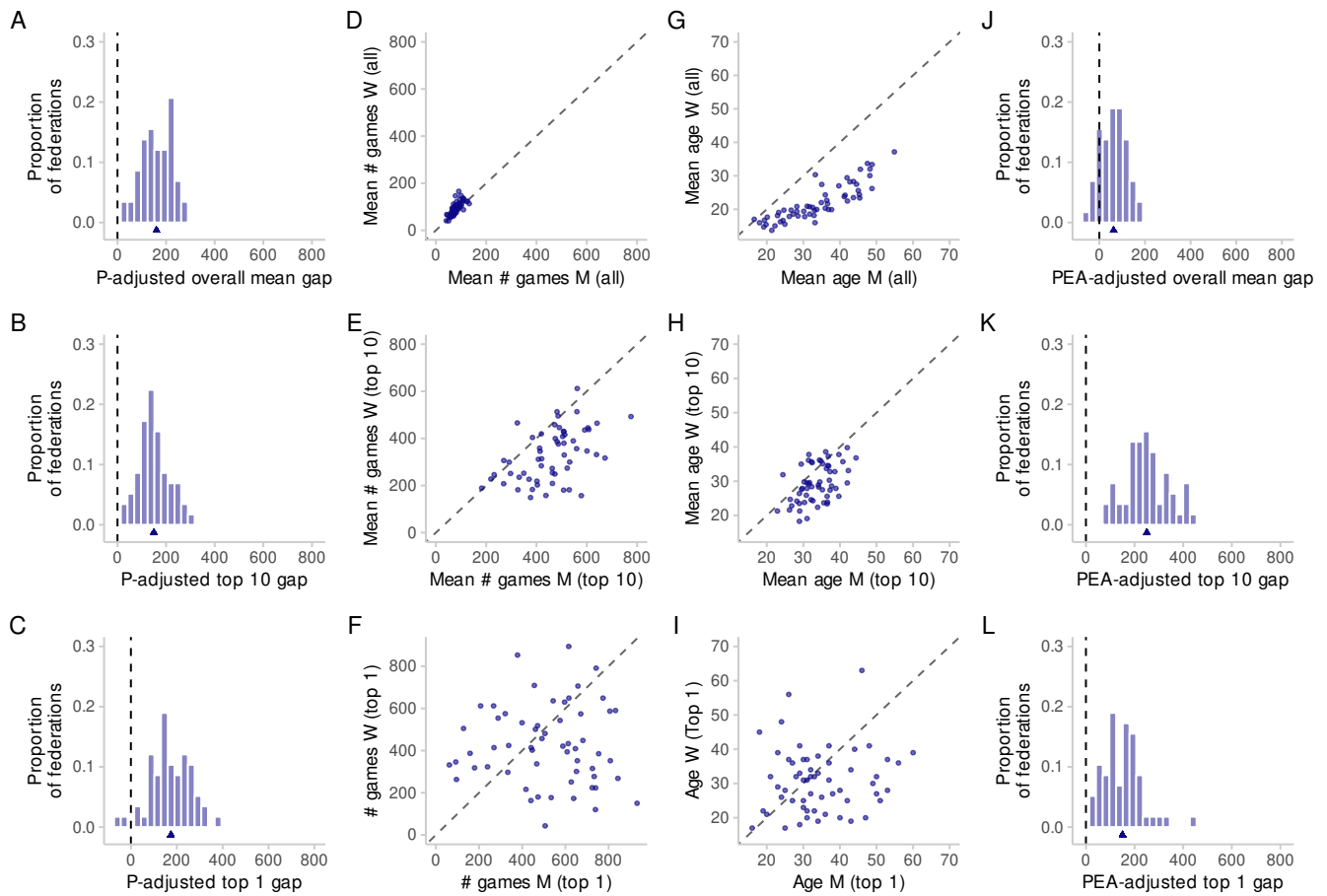


Fig. S1. As Fig. 3 in the main text, but with a rating floor of 1400 (sample size in each panel: 58).

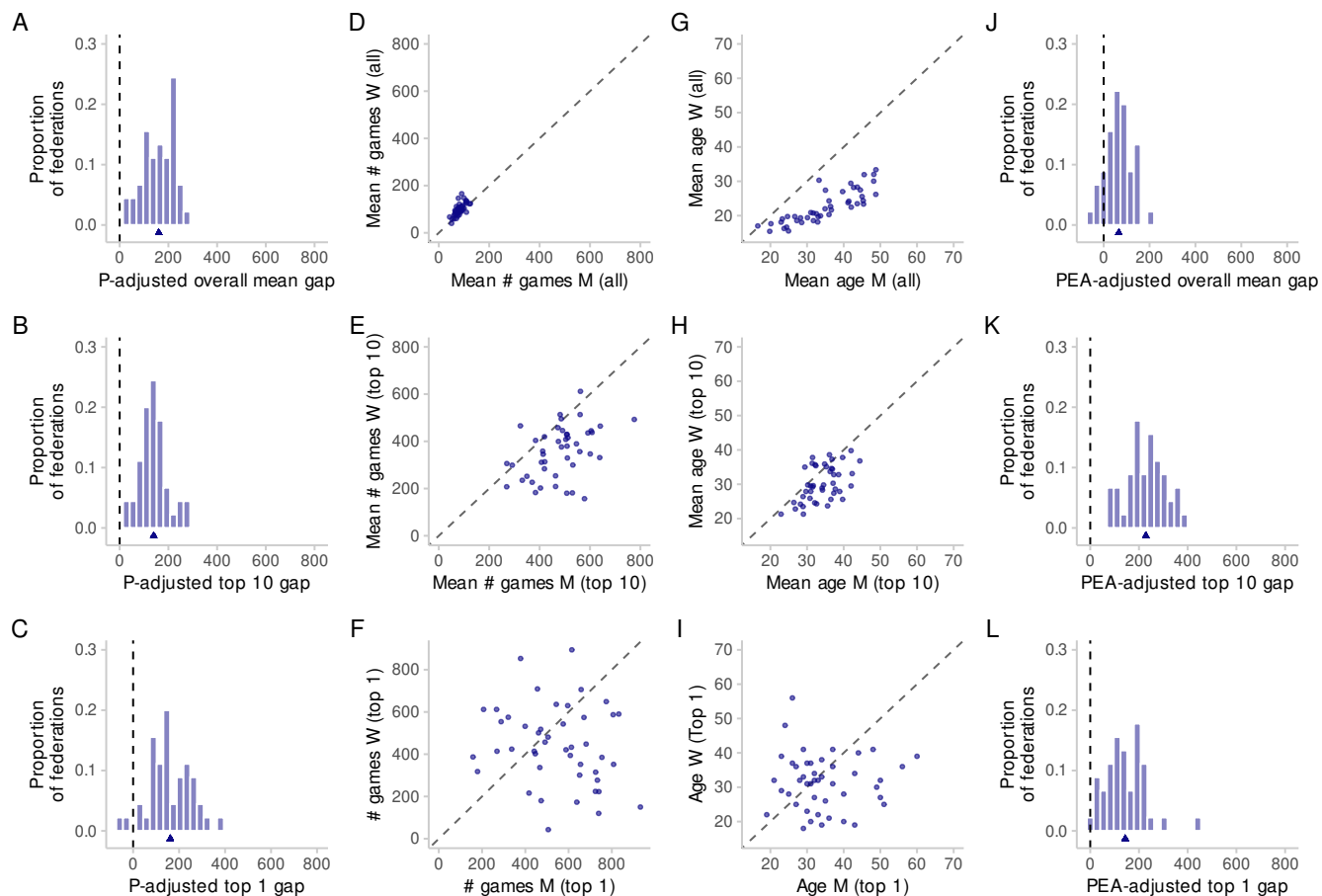


Fig. S2. As Fig. 3 in the main text, but with a rating floor of 1600 (sample size in each panel: 45).

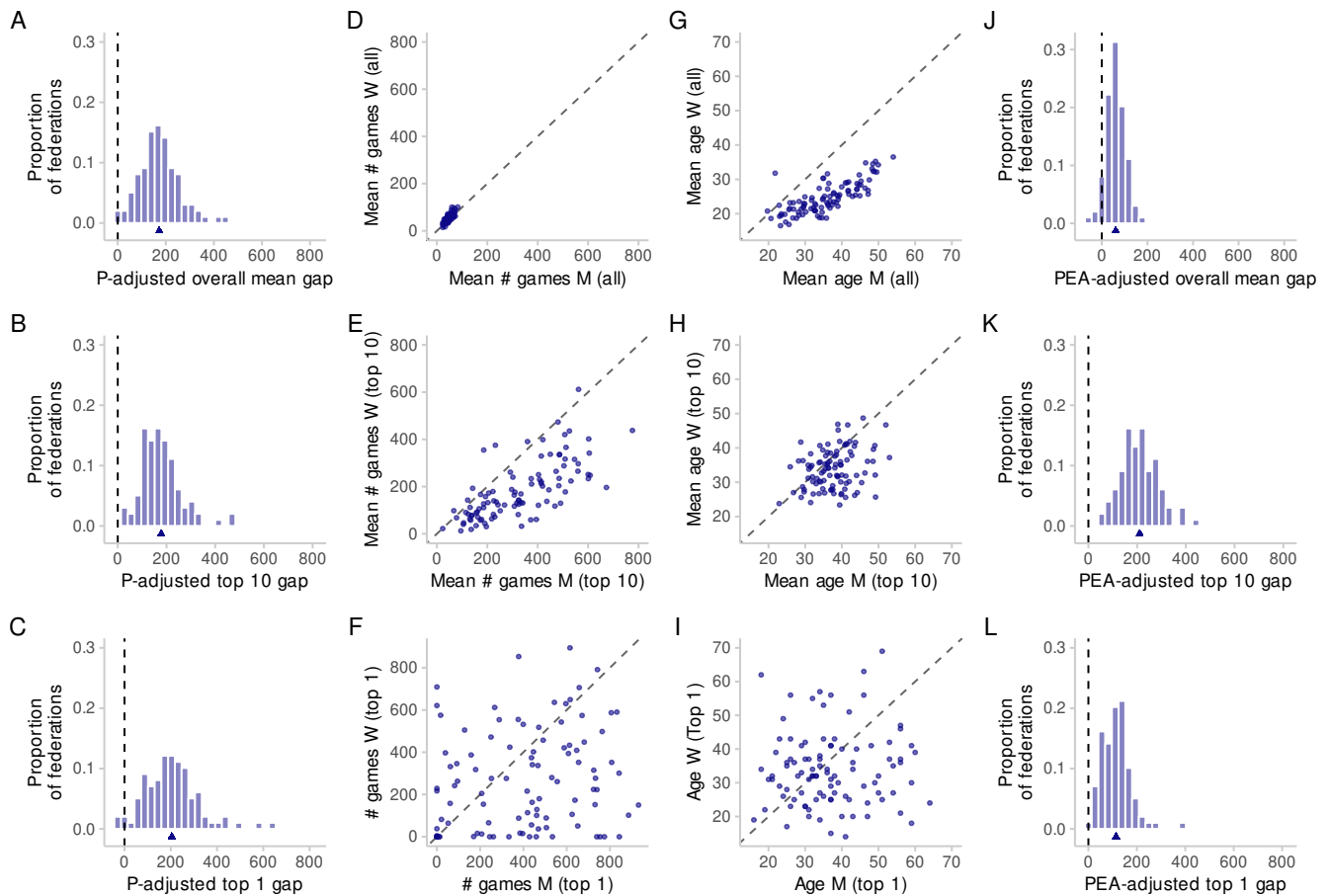


Fig. S3. As Fig. 3 in the main text, but with both junior and inactive players, and a rating floor of 1000 (sample size in each panel: 99).

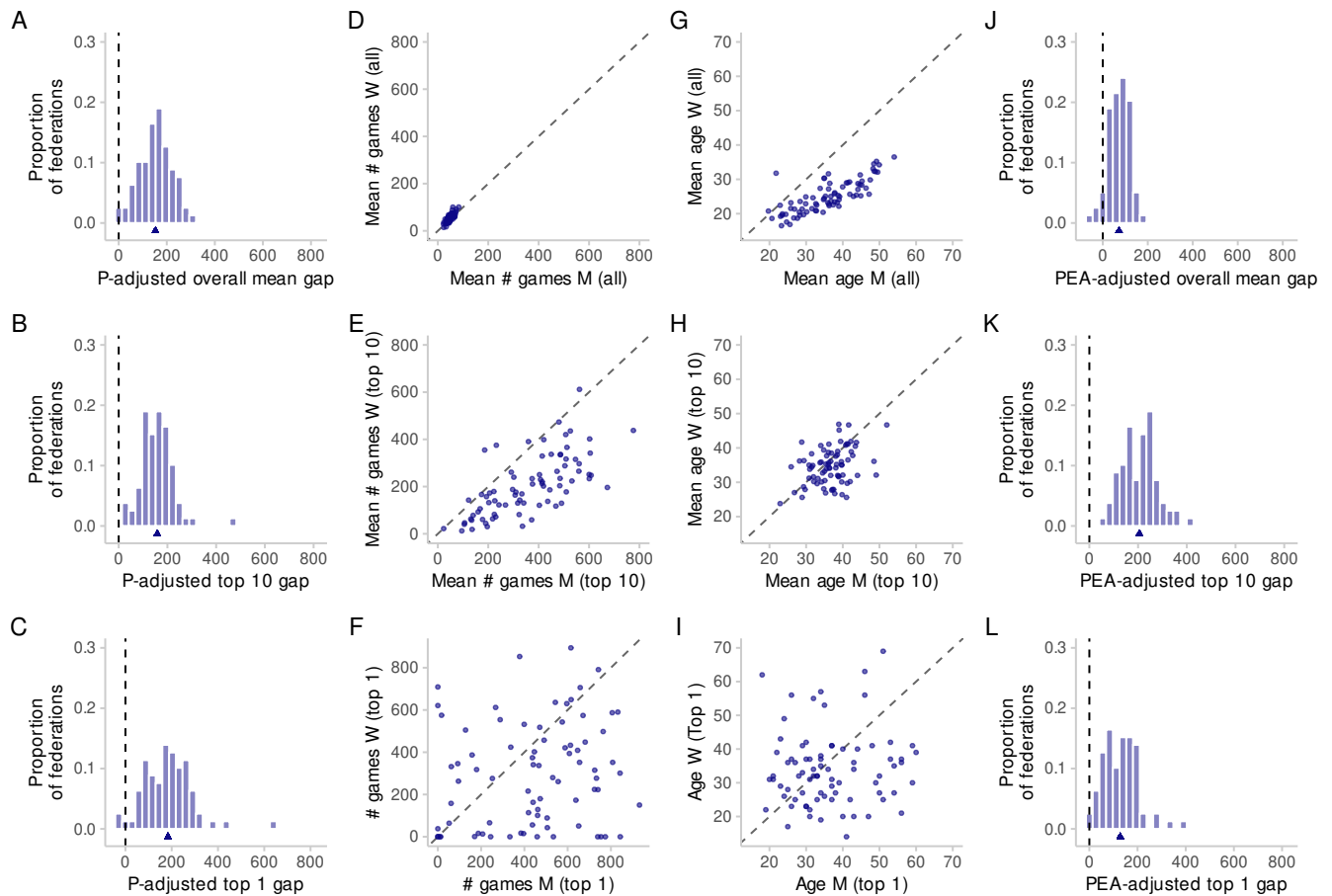


Fig. S4. As Fig. 3 in the main text, but with both junior and inactive players, and a rating floor of 1400 (sample size in each panel: 79).

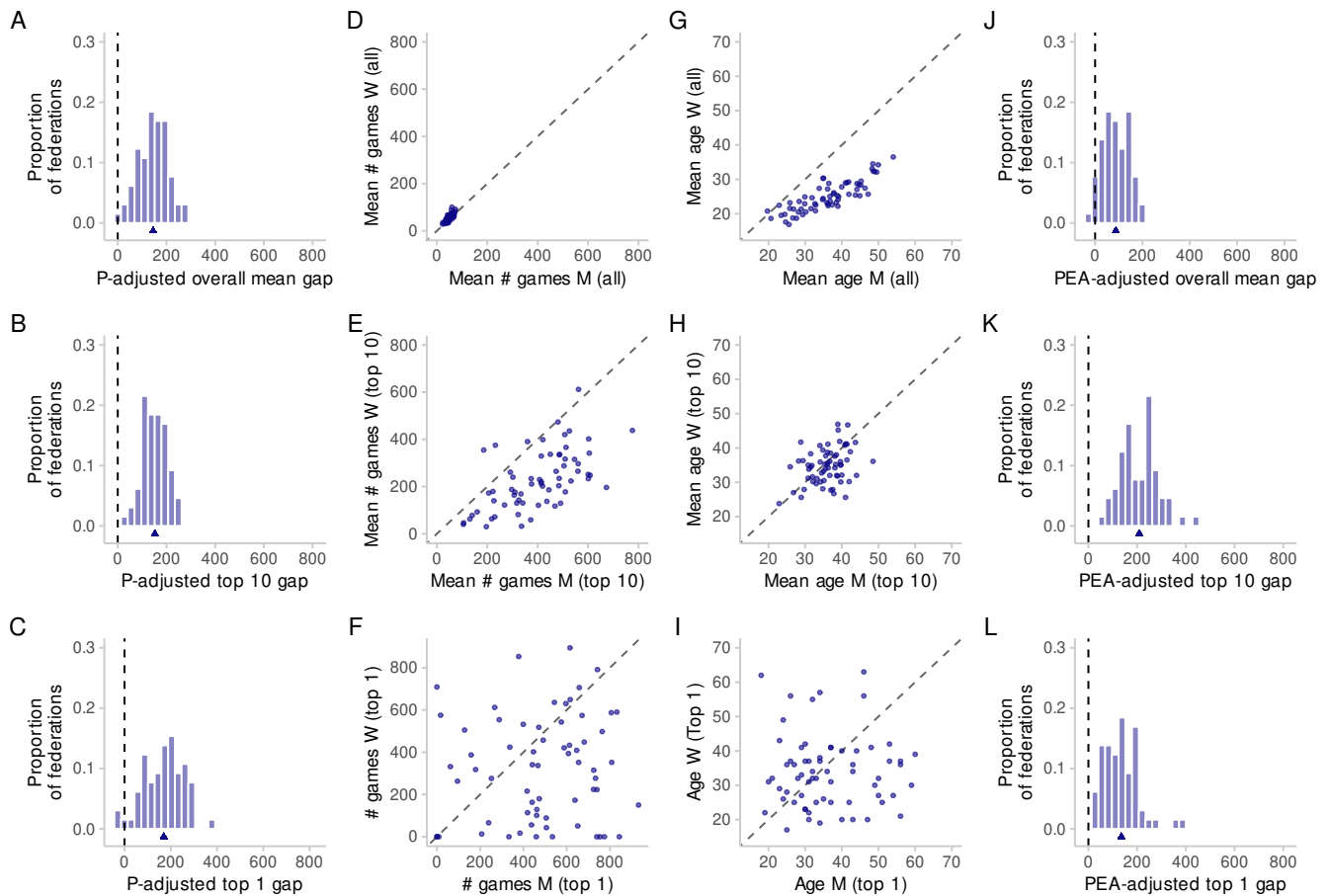


Fig. S5. As Fig. 3 in the main text, but with both junior and inactive players, and a rating floor of 1600 (sample size in each panel: 65).

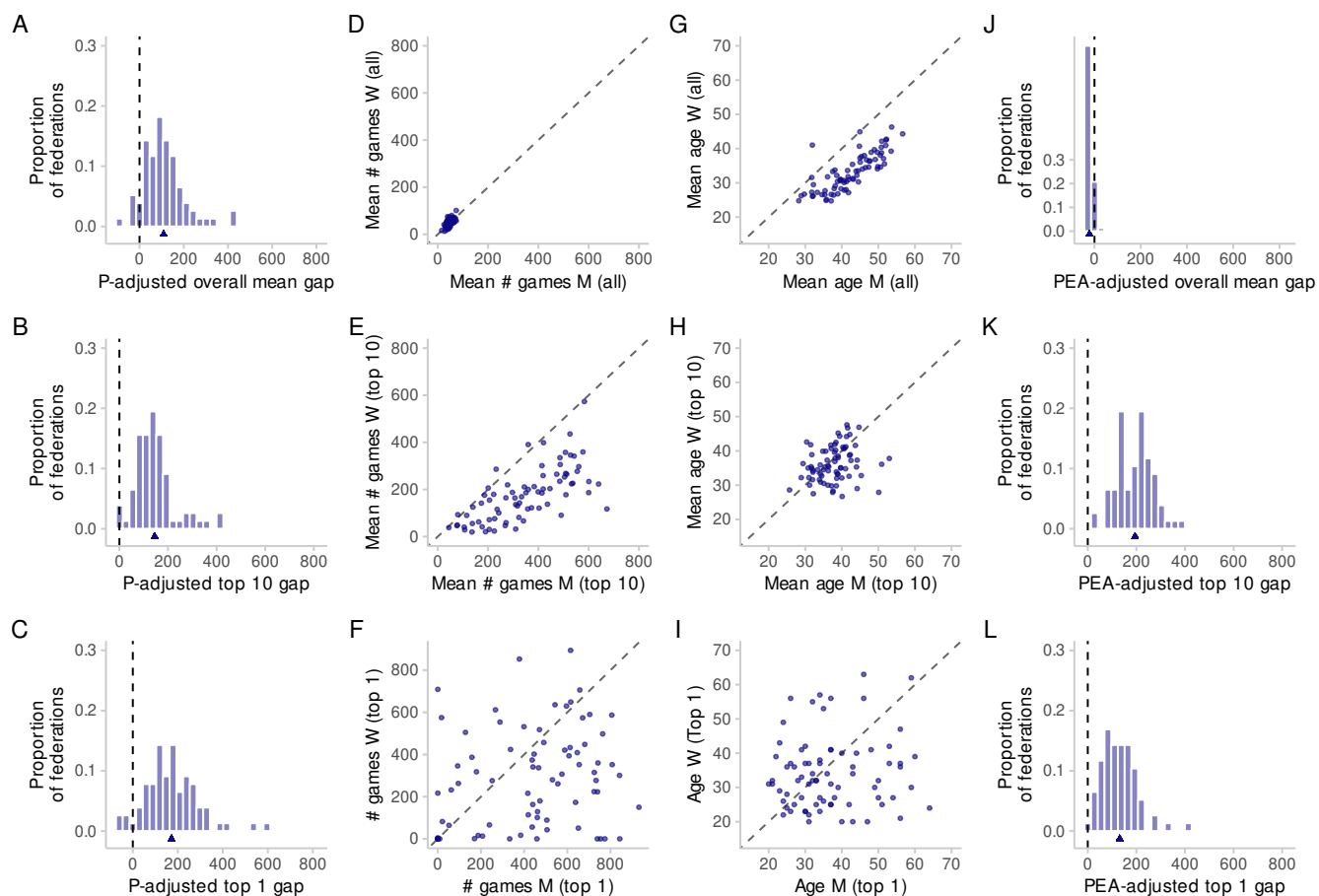


Fig. S6. As Fig. 3 in the main text, but without junior and with inactive players, and a rating floor of 1000 (sample size in each panel: 77).

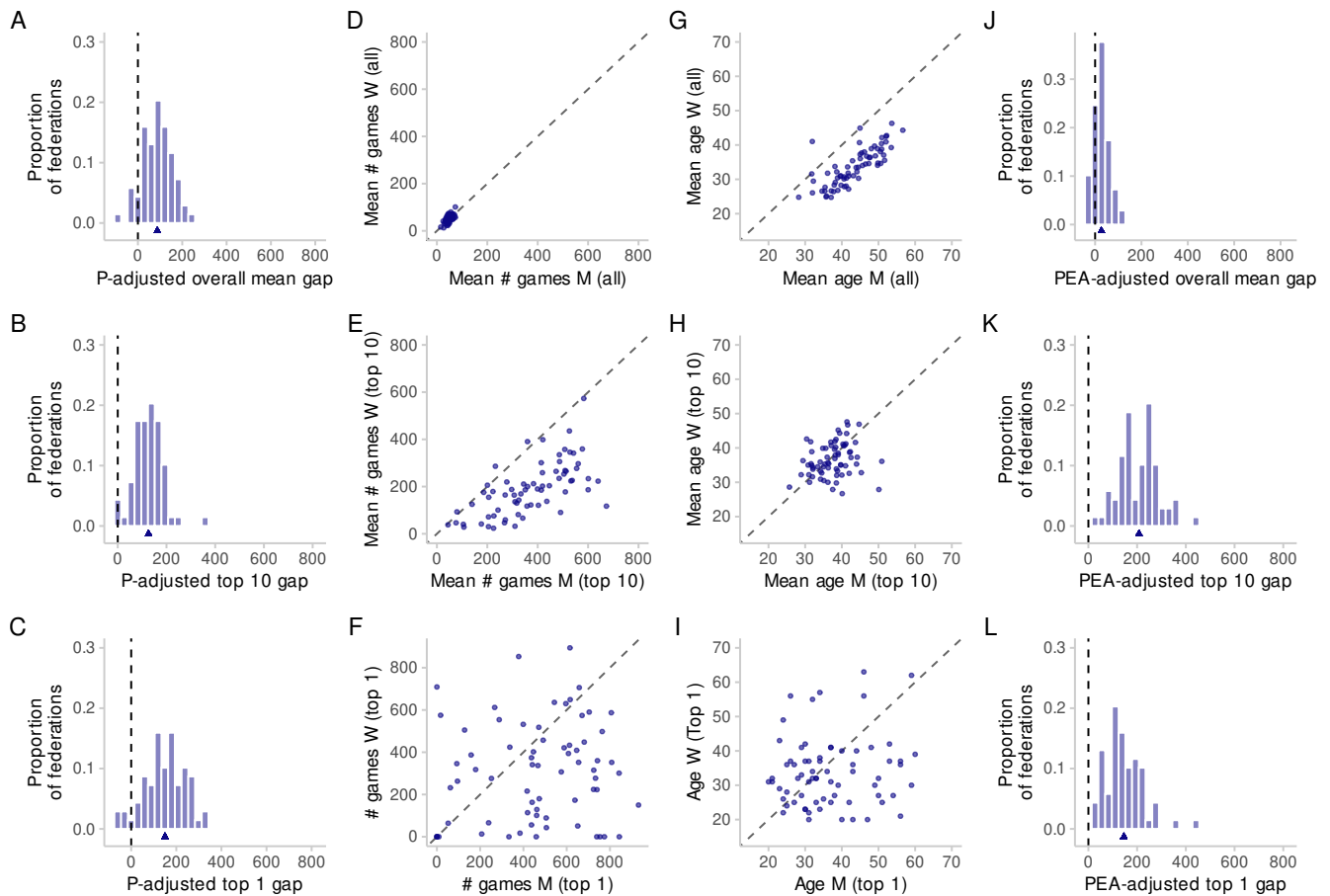


Fig. S7. As Fig. 3 in the main text, but without junior and with inactive players, and a rating floor of 1400 (sample size in each panel: 69).

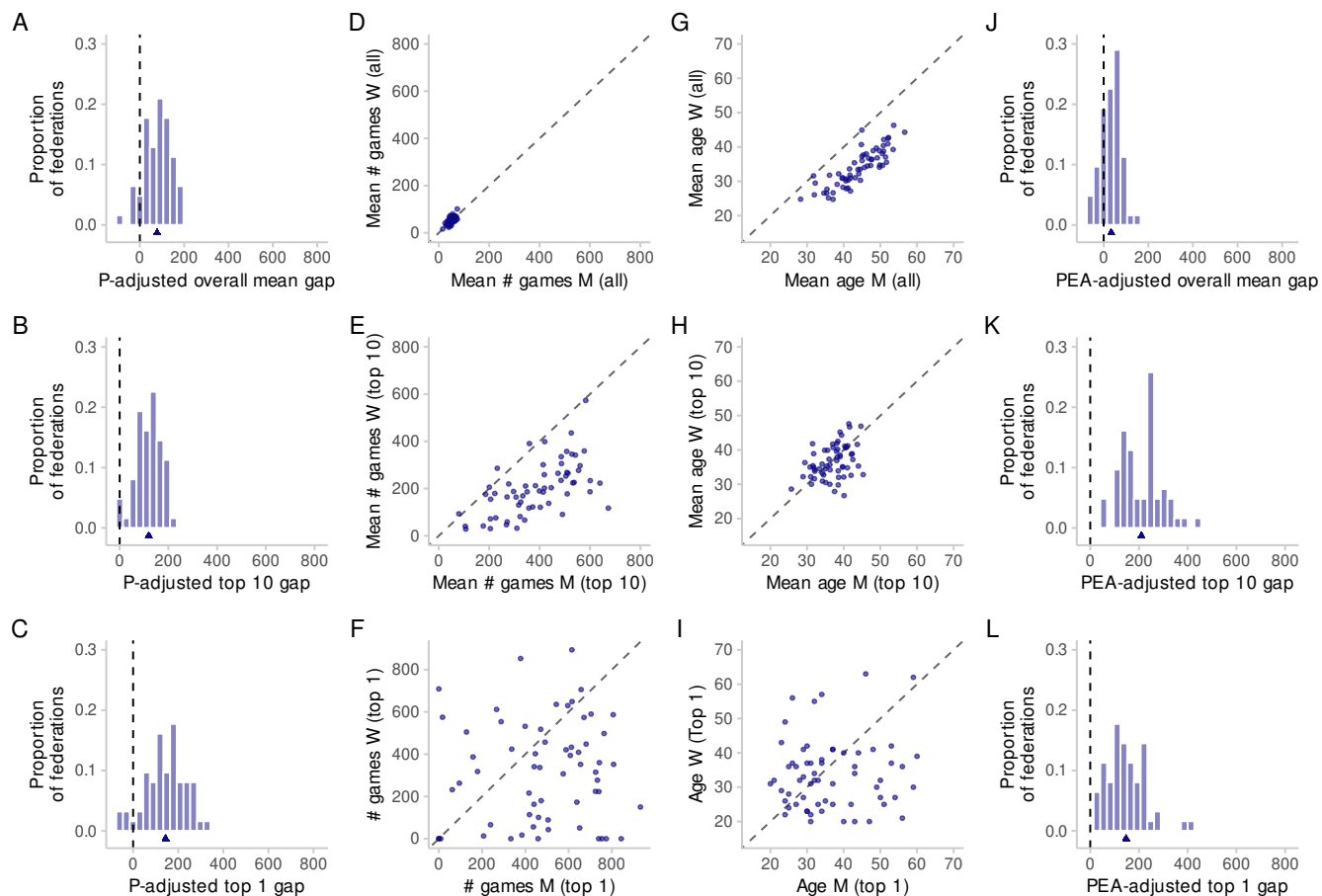


Fig. S8. As Fig. 3 in the main text, but without junior and with inactive players, and a rating floor of 1600 (sample size in each panel: 62).

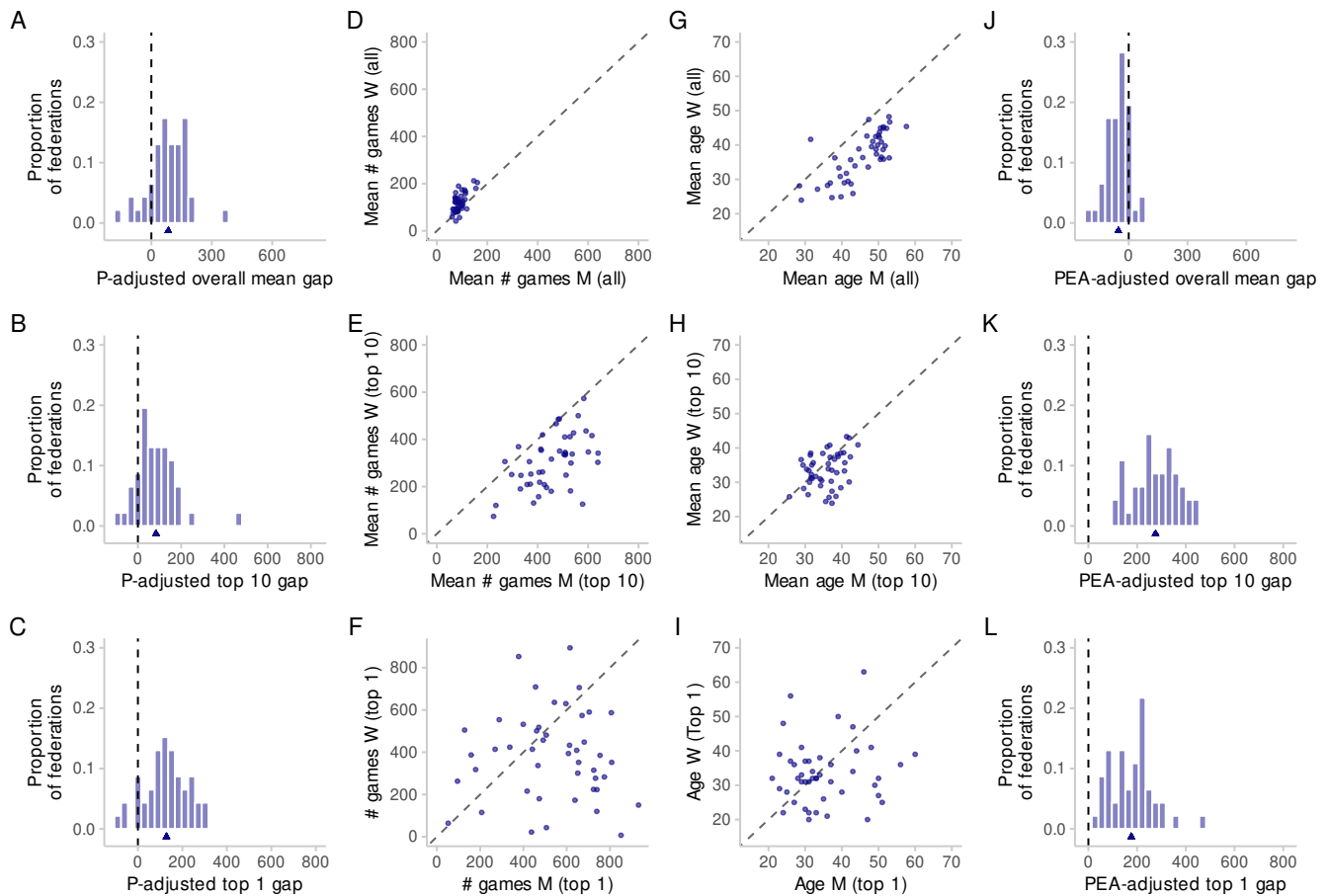


Fig. S9. As Fig. 3 in the main text, but without junior and without inactive players, and a rating floor of 1000 (sample size in each panel: 46).

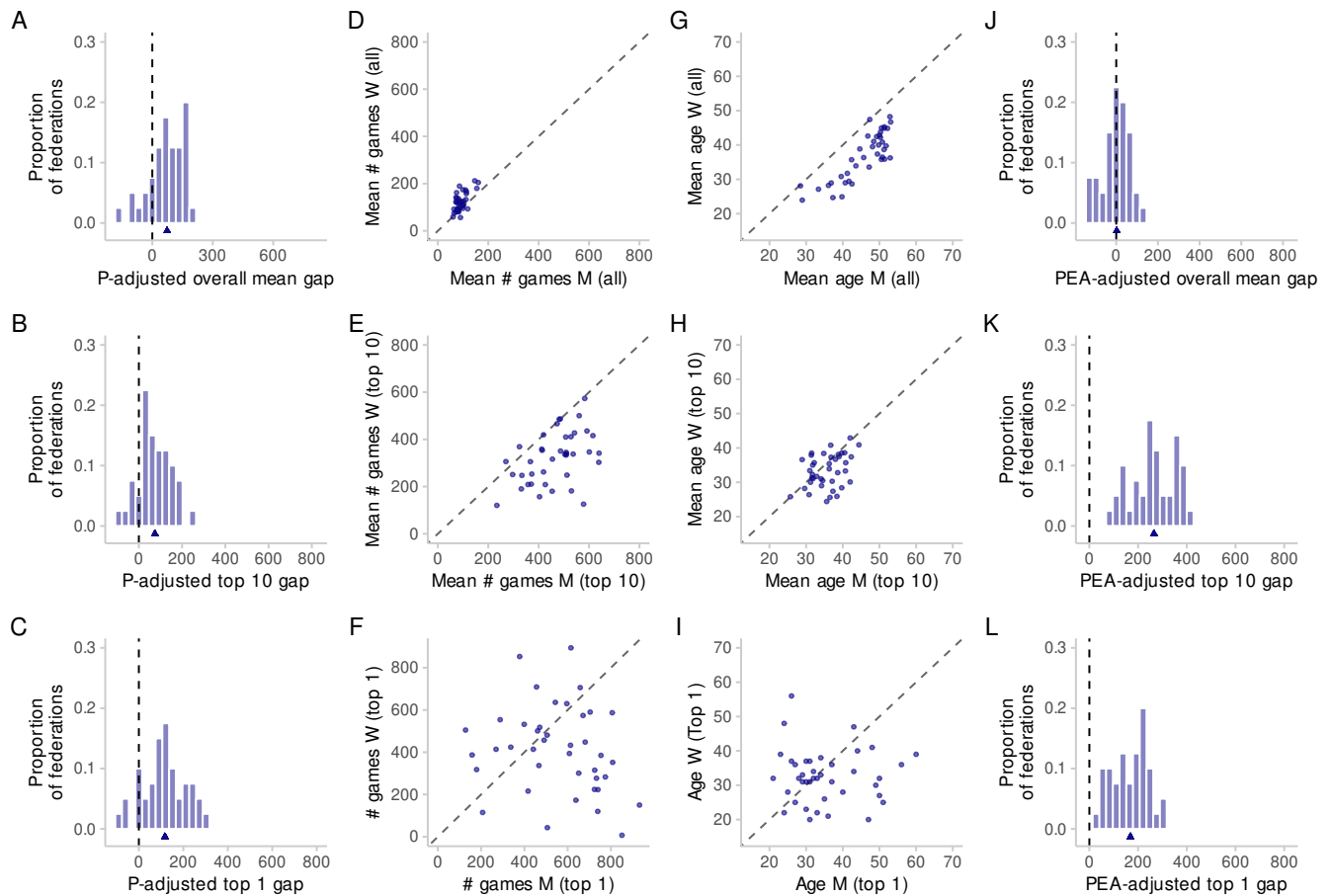


Fig. S10. As Fig. 3 in the main text, but without junior and without inactive players, and a rating floor of 1400 (sample size in each panel: 40).

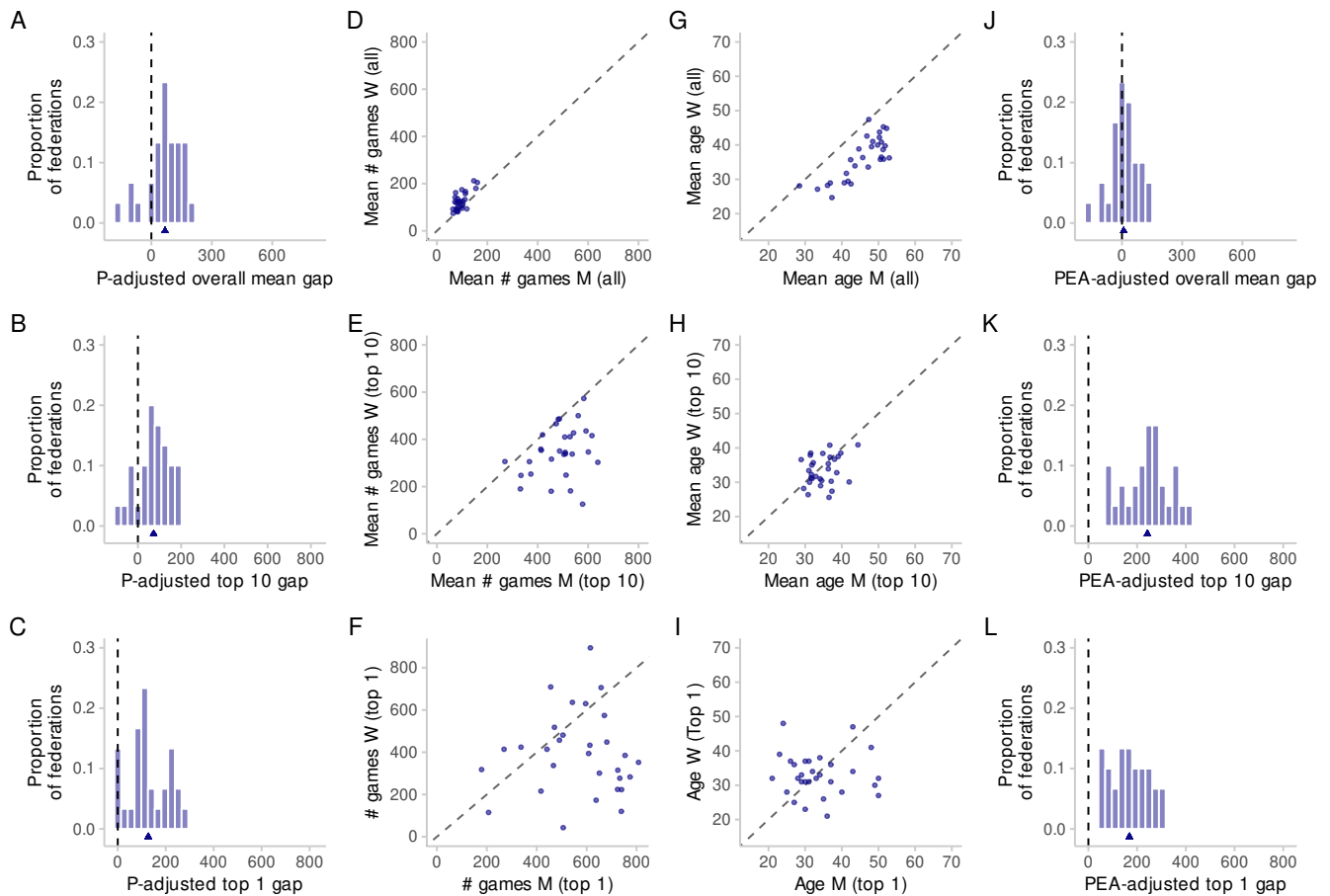


Fig. S11. As Fig. 3 in the main text, but without junior and without inactive players, and a rating floor of 1600 (sample size in each panel: 30).

52 **References**

- 53 1. M Knapp, Are participation rates sufficient to explain gender differences in chess performance? *Proc. Royal Soc. B: Biol.*
54 *Sci.* **277**, 2269–2270, doi:10.1098/rspb.2009.2257 (2010).